
Zero-Inflated Beta, Tobit, And Latent Growth Mixture Models For Analyzing GenAI Usage Trends And Trajectories

Jerahmeel S. Hipolito^{1,*}, Angeli D. Bergonio¹, Carina Jean A. Damasco¹, Lance Aron O. Javier¹, Dr. Marco C. Mandap¹

¹*Department, College of Science, Bulacan State University, City of Malolos, Bulacan 3000, Philippines*

ABSTRACT

With the rise of GenAI, its increasing use in academia has become a cause for concern regarding research integrity. However, studies analyzing its usage trends, whether in global or local contexts and even in Bulacan State University (BulSU), are still limited. This study analyzes GenAI usage trends in BulSU from 2013 to 2023, analyzing 6,343 research papers authored by students and detected 262 AI-flagged papers using Quillbot Premium. Due to the presence of excess zeroes in the data, Zero-Inflated Beta and Tobit models were applied to trend analysis, while Latent Growth Mixture Models (LGMM) determined distinct class trajectories. By 2023, the expected AI scores of Quadratic Tobit were censored at 0% for individual-level and reached 33.36% for discipline-level. Additionally, the trend of AI scores behaves in a way that suggests a rapid move toward full adoption (censored at 100%) by 2026 for both levels, based on projections extrapolated from the 2013–2023 observed data. Furthermore, the Quadratic LGMM classified disciplines into three latent class groups and are described based on Roger's Innovation Diffusion Theory with Innovators and Early Adopters, Early and Late Majority, and Laggards, where Early and Late Majority show a trend behavior that reaches GenAI usage censored at 100% by year 2026, which overtakes Innovators and Early Adopters with only 87.33%. The findings show that GenAI usage in academic research shows a clear upward trend while varying from disciplines. As GenAI may become fully integrated in the coming years based on the direction of the current trend, the study highlights the need for ethical GenAI usage while encouraging innovation in research.

Keywords: GenAI trends and trajectories, GenAI usage and adoption; Latent Growth Mixture Model; Tobit; Zero-Inflated Beta

***Corresponding author:** Jerahmeel S. Hipolito, Mathematics Department, College of Science, Bulacan State University, Philippines. E-mail: hipolitojerahmeel12@gmail.com

1. Introduction

As Artificial Intelligence (AI) advances, Generative AI (GenAI) has disrupted conventional processes, sparking a revolution in content creation, research, and automation across fields. This transformation is evident in how people interact with technology as GenAI's capabilities expand from simple text generation to more complex tasks like data analysis, content creation, research assistance, and productivity enhancement [1]. The utilization of GenAI has caused institutions and industries to reconsider traditional methods. Some are adopting new methods which are driven by the enhanced capabilities of AI, while others are still skeptical of this new advancement. Such change brings significant effects, modifying processes, and sets new benchmarks for efficiency and innovation. By learning from existing data, GenAI can generate new, meaningful content like text, images, or audio. These systems have the capabilities to create artistic works, such as text which mimics writers, or images that resemble the styles of illustrators while also serving as intelligent tools that are capable of understanding and responding to complex questions [2]. This transformation especially impacts the academia, where GenAI supports various research tasks.

The rise in the capabilities of GenAI aligns with its increasing usage in academia, by being employed for a variety of tasks such as automating literature reviews and creating content [3]. As AI technology evolves, its academic impact has become necessary, transforming areas like data analysis and content generation, which significantly enhance research productivity [4]. A crucial point to the transformation was the launch of GPT-3 in 2020, where it introduced the capability to produce coherent and contextually relevant text, which makes it a valuable tool for researchers [5]. Although this trend improves efficiency, it still raises issues about its potential impact on

originality, critical thinking, and academic integrity [6, 1].

As GenAI continues to evolve across various fields, particularly in academia, several studies have analyzed its growing influence on academic writing. Kobak et al. [7] suggest that analyzing GenAI usage trends can be summarized into three main approaches: detector tools trained on human and machine-generated texts, mixture models of word frequencies, and the analysis of specific "marker words" commonly overused by large language models (LLMs). These methods help quantify AI involvement.

While most of the studies [7-9] primarily utilize mixture models of word frequency and analysis of "marker words" to analyze GenAI usage trends, recent but limited studies have placed greater emphasis on detecting GenAI usage trends using AI detection tools. An example is a study [10] where they employed the Binoculars score, which is a statistical feature-based classifier to detect LLM-generated content in preprint platforms such as arXiv, medRxiv, and bioRxiv, which reveals a significant rise in AI-assisted writing after ChatGPT's post-launch. Another study [11] utilized the Originality.AI detection tool in the analysis of submissions on arXiv. They identified a significant increase in papers written by AI in computer science, having an increase usage from 3.17% in 2019 to 7.37% by 2023. Similarly, one study [12] used tools such as GPTKit and Smodin when analyzing AI-written probabilities within arXiv abstracts and recorded increased usage by non-native speakers of English. Although its developments are promising, existing research that employ AI detection tools to quantify GenAI usage remains limited and to date, no studies have applied advanced statistical modeling to analyze GenAI usage trend comprehensively, which highlights a significant gap as to how GenAI usage trends

and growth trajectories are currently understood.

Although the global attention to GenAI is significantly growing, there are still limited existing studies in Philippine academic context. Most studies rely on self-reported data to assess perceptions and attitudes [13, 14]. These lack the empirical analysis that is necessary to measure actual usage. Although trend analysis has been applied in other fields [15], no study in the country has used advanced statistical methods such as Zero-Inflated Beta, Tobit, or Latent Growth Mixture Models (LGMM) to track GenAI usage trends, methods well-suited for datasets with excess zeros and censored values. In addition, fewer than twenty AI-related studies exist at Bulacan State University (BulSU) based on BulSU E-Library, with most of them focusing on Information Technology development. Also, when it comes to studies on GenAI at BulSU, the only notable study is the one that investigated the intention to use and acceptance of GenAI among college students and educators using the Unified Theory of Acceptance and Use of Technology (UTAUT) framework [16]. While that study reported high usage rates, it focused on intention and perception rather than empirical usage trends. This reveals a critical gap, both nationally and at BulSU, in applying data-driven methods to examine the trends and growth trajectories of GenAI usage in academic research.

With that, this study aims to explore and analyze the trends and growth trajectories of GenAI usage in academic research papers at Bulacan State University from 2013 to 2023. While international studies that use AI detection tools in analyzing GenAI usage trends are existing, they are still relatively limited compared to more common approaches like mixture models of word frequencies and use of specific “marker words.” In the Philippines however, application of AI and plagiarism

detection tools for empirical trend analysis are still novel, which makes this study one of the first locally grounded studies to adopt such method. These prior approaches often depend on surface-level linguistic patterns and lack the integration of advanced statistical modeling which in contrast, this study introduces robust methods such as Zero-Inflated Beta (ZIB), Tobit, and Latent Growth Mixture Model (LGMM) in capturing deeper patterns in the data. Aside from uncovering the overall trend in GenAI usage, LGMM will also identify the distinct latent class of research behavior and its unique growth trajectories over time. This comprehensive data-driven approach will close a significant gap in the literature, providing new insights on the empirical study of GenAI usage in Philippine academia, and offer evidence-based recommendations for institutional policies at BulSU regarding the ethical effective application of GenAI in academic research.

1.1 Research Objectives

In recent years, there has been a significant increase in the usage of Generative Artificial Intelligence (GenAI) tools in academia, which indicated a growing trend in the integration of technology into educational practices. With that, this study aims to explore the trends and growth trajectories of GenAI usage over time at Bulacan State University (BulSU), providing insights into how this technology is shaping the academia.

Specifically, the researchers seek to answer the following questions:

1. How has GenAI usage in research papers evolved from 2013 to 2023, specifically in terms of trends and growth trajectories for AI Score at: (a) individual-level paper; and (b) discipline-level paper?
2. How does the Plagiarism Score influence the trends and growth patterns of AI Score in research papers from 2013 to 2023 at discipline-level?

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3. How many distinct latent growth classes can disciplines be categorized based on their GenAI usage, and what are the characteristics of these classes?
 4. Do the aggregate trends of AI Score in research papers correspond with latent class growth trajectories identified in Latent Growth Mixture Model?
 5. Which trend analysis model (Zero-Inflated Beta and Tobit Variants) and which Latent Growth Mixture Model approach (Quadratic and Piecewise) best captures AI Score at the individual and discipline levels?
 6. Based on the best trend analysis model and the best Latent Growth Mixture Model approach, how will the AI Score be expected to behave beyond 2023 at both the individual and discipline levels?
 7. What policies and recommendations can be derived from the study to guide BulSU and other institutions in managing GenAI use in research and education?

2. Materials and Methods

This section presents the research methodology, sources of data, and data analysis used to address the problem of the study. This section provides a framework for understanding the approach in exploring the trends and growth trajectories of GenAI Usage in academia from 2013 to 2023.

2.1 Research Design

This study utilized a Quantitative Descriptive Research Design and a Longitudinal Research Design, specifically the trend analysis models which are the Zero-Inflated Beta (ZIB) and Tobit Model, and Latent Growth Mixture Model (LGMM), in exploring the trends and trajectories associated with Generative AI (GenAI) usage in research papers from 2013 to 2023. Quantitative Descriptive Research Design involves describing individuals, events, or conditions through a structured analysis of their characteristics [17]. Additionally, a Longitudinal Research Design involves

following subjects over a defined period and collecting data at specific intervals [18]. This design is effective for observing changes in variables over time, making it ideal for assessing the progression of GenAI usage within the designated period.

2.2 Sources of Data

The study utilized secondary data collected from the Bulacan State University (BulSU) repository, accessed through the E-Library system of the university. The data consists of research papers from 2013 to 2023, which allows for an in-depth analysis of GenAI usage trends and trajectories over time. The data collection process involved making formal requests and accessing information directly online. A formal request was submitted to the BulSU E-Library to gain access to the research papers. Upon retrieval, the data underwent extensive cleaning and organization through MS Excel and normalization through MS Access. Furthermore, formal interviews were conducted with BulSU E-Library staff to gain insights into the management processes of the repository.

2.3 Data Analysis

In addressing the research questions, different quantitative analyses were used to describe and examine the trends and growth trajectories of GenAI use over time. The independent variable was the period between 2013 and 2023, while the dependent variable was the GenAI use in research articles, measured in terms of AI score. Furthermore, plagiarism score was considered as a time-varying covariate to evaluate its effect on the trajectory of GenAI Usage. The study also examined how aggregate trends aligned with individual growth trajectories and offered suggestions for developing institutional policies.

2.3.1 Data Pre-processing

For data accuracy and integrity, comprehensive data preprocessing was conducted. The research papers obtained

were first cleaned and organized through MS Excel. The whole process contained addressing missing values, removing duplicates, and structuring data fields for consistency. Additionally, a data normalization up to second normal form (2NF) was performed using the MS Access software to ensure the elimination of redundancy and dependency issues through normalization, especially in the repositories of Bulacan States University, to facilitate efficient data queries and analysis. These preprocessing steps were essential in conducting trend analysis and latent growth mixture modeling.

After data normalization, querying the database by discipline (based on the Library of Congress Classification system) provided convenience for end-users. Descriptive statistics were then used to compute counts, percentages, and visualize the frequencies of research papers per discipline and per year from 2013 to 2023.

2.3.2 Nature of Dataset

The dataset consists of AI and plagiarism scores assigned to research papers published between 2013 and 2023, ranging from 0 to 1, generated by Quillbot Premium. A key characteristic of the dataset is the high presence of zero AI scores, indicating minimal GenAI usage in earlier years, followed by a sharp increase post-pandemic. Due to this excess of zeros, specialized models such as the Zero-Inflated Beta (ZIB) Model and Tobit Model were used to accurately assess GenAI usage patterns. This initial exploration provided insights into the evolution of AI-related research across disciplines.

To analyze AI trends at different levels, two forms of the dependent variable are considered. First is the individual-level paper where the AI score y_{it} represents the AI score of an individual paper i for $i = 1, 2, 3, \dots, N$ at year t for $t = 0, 1, 2, 3, \dots, 10$ from 2013 to 2023 for $0 \leq y_{it} \leq 1$. This

measure is used in trend analysis models (ZIB and Tobit Model) to assess GenAI usage at the individual-level. The second is the discipline level paper y_{agg} , which aggregates AI scores or y_{it} for each discipline j at year t to observe overall trends. This aggregated AI score for discipline j at year t is calculated as:

$$y_{agg} = \frac{\sum_{i=1}^{n_{jt}^{AI}} y_{it}}{n_{jt}^{AI}}, \quad 0 \leq y_{agg} \leq 1 \quad (2.1)$$

where y_{it} is AI score of paper i at year t , and n_{jt}^{AI} is number of papers in discipline j at year t given the paper is flagged as AI-generated.

However, some disciplines have a limited number of AI-flagged papers in specific years, which may introduce bias in aggregation. To address this, a weighting factor is applied to prevent disciplines with small sample sizes from disproportionately influencing the results. The sampling weight is defined as:

$$w_{jt} = \frac{1}{n_{jt}} \quad (2.2)$$

where n_{jt} = number of papers in discipline j at year t . This adjustment accounts for variations in sample sizes across disciplines and years. With that, the final weighted aggregated AI score is then calculated as:

$$y_{agg, w_{jt}} = y_{agg} \cdot w_{jt}, \quad 0 \leq y_{agg, w_{jt}} \leq 1 \quad (2.3)$$

This adjustment ensures that variations in sample sizes across disciplines are accounted for, making the aggregated measure more representative of GenAI usage trends. This measure is applied in trend analysis for ZIB and Tobit models, and for Latent Growth Mixture Modeling to capture the evolution of AI usage across disciplines over time.

2.3.3 Trend Analysis

To understand the general patterns in GenAI usage, a trend analysis is conducted. Since Generative AI (GenAI) usage in research papers is relatively new, trend analysis helps identify when and how GenAI usage increased, as well as whether certain

disciplines use GenAI earlier than others. However, the presence of high zero AI scores where most papers showed no AI content detected, have made traditional statistical methods like linear regression not suitable for modeling these trends. In addressing this, the study conducted two statistical model approaches, the Zero-Inflated Beta (ZIB), and Tobit Model. These methods can make a more accurate representation of AI usage trends by accounting for the large amount of zero AI scores and the bounded nature of AI detection results.

The first part is Zero-Inflated Beta (ZIB) Regression, a statistical model used for datasets with a high number of zero values and continuous proportions between 0 and 1 [19]. In this study, AI scores in research papers exhibit this characteristic, where most papers show zero AI usage, while others contain AI-generated content with varying proportions.

The ZIB model addresses this by splitting the data into two processes: (1) a logistic regression component which models the probability that a paper has an AI score of zero, and (2) a beta regression component which models AI scores for papers with detected AI content [20, 21]. This dual structure improves the estimation of AI adoption trends by capturing both the presence and extent of AI use. The ZIB model is especially useful when analyzing bounded continuous data with many zeros, allowing differentiation between true non-users of AI (papers with zero AI usage) and low AI score cases (papers with minimal AI content).

With that, the probability of a research paper having a zero AI score for the individual-level shown in eq. (2.4) and discipline-level shown in eq. (2.5) model can be expressed based on a previous study [22] as:

$$\text{logit}(v_{it}) = \frac{v_{it}}{1-v_{it}} = \alpha_0 + a_i \quad (2.4)$$

$$\text{logit}(v_{jt}) = \frac{v_{jt}}{1-v_{jt}} = \alpha_0 + \alpha X_{agg,w_{jt}} + a_i \quad (2.5)$$

where v_{it} is the probability that paper i at year t has a zero AI score, α_0 is the intercept term, a_i is the individual-specific random intercepts, and $X_{agg,w_{jt}}$ is the aggregated plagiarism score as time-varying covariate.

In addition, for nonzero AI scores, the expected AI score is modeled using Beta regression. Based on the same work [22], the logit-transformed mean AI score for the individual-level shown in eq. (2.6) and discipline-level shown in eq. (2.7) model is expressed as:

$$\text{logit}(\mu_{it}) = \frac{\mu_{it}}{1-\mu_{it}} = \beta_0 + b_i \quad (2.6)$$

$$\text{logit}(\mu_{jt}) = \frac{\mu_{jt}}{1-\mu_{jt}} = \beta_0 + \beta X_{agg,w_{jt}} + b_i \quad (2.7)$$

where μ_{it} is the expected AI score of paper i at year t , β_0 is the intercept term, b_i is the individual-specific random intercepts, and $X_{agg,w_{jt}}$ is the aggregated plagiarism score as time-varying covariate.

Lastly, The ZIB model was implemented using the *gamlss* package in R and *ggplot2* for visualization.

Moving forward, the second part is the Tobit model, a statistical approach designed for situations where the dependent variable is censored at a certain threshold [23]. In the context of this study, the Tobit model was applied to analyze trends in GenAI usage, as many research papers exhibited zero AI scores, making a censored regression approach appropriate for modeling the underlying relationships. With that, the Tobit model is applied in three forms: the Classic or Linear Tobit Model, the Quadratic Tobit Model, and the Piecewise Tobit Model. Each specification addresses different structural properties in the data, allowing for a more refined analysis of GenAI trends across research papers.

In this model, AI scores y_{it} are treated as latent continuous variables that are only observed if they fall within the range defined by the lower limit L and upper limit U . If the true AI score is below or above these bounds,

it is recorded as left-censored or right-censored, respectively. This structure allows for a more accurate representation of AI adoption trends in research papers, particularly in capturing the differences between non-users of AI (zero scores) and varying levels of AI usage (continuous scores between 0 and 1) [24].

With that, the latent Tobit Model equation for the individual-level shown in eq. (2.8) and discipline-level shown in eq. (2.9) is expressed as:

$$\begin{aligned} y_{it}^* &= \beta_0 + \beta_1 x_{it} + \epsilon_{it}, \quad \epsilon_{it} \sim N(0, \sigma^2) \quad (2.8) \\ y_{agg, w_{jt}}^* &= \beta_0 + \beta_1 x_{it} + \beta_2 X_{agg, w_{jt}} + \epsilon_{it}, \\ &\quad \epsilon_{it} \sim N(0, \sigma^2) \quad (2.9) \end{aligned}$$

where y_{it}^* is the latent (unobserved) AI score y_{it} of paper i at year t , β_0 is the intercept term, β_1 is the coefficient for time trend variable x_{it} , β_2 is the coefficient for time-varying covariate $X_{agg, w_{jt}}$, x_{it} is the time index of paper i at year t from 2013 to 2023, and ϵ_{it} is the latent (unobserved) error term of paper i at year t . Here, the latent AI score y_{it}^* and $y_{agg, w_{jt}}^*$ is only observed if it exceeds zero, and the error term assumed to follow a normal distribution with mean zero and variance σ^2 .

In addition, the observed AI score y_{it} and $y_{agg, w_{jt}}$ follows the censoring rule of:

$$y_{it} = \begin{cases} L, & \text{if } y_{it}^* \leq L \\ y_{it}^*, & \text{if } L < y_{it}^* < U \\ U, & \text{if } y_{it}^* \geq U \end{cases} \quad (2.10)$$

$$y_{agg, w_{jt}} = \begin{cases} L, & \text{if } y_{agg, w_{jt}}^* \leq L \\ y_{agg, w_{jt}}^*, & \text{if } L < y_{agg, w_{jt}}^* < U \\ U, & \text{if } y_{agg, w_{jt}}^* \geq U \end{cases} \quad (2.11)$$

This ensures that any negative latent AI score is recorded as zero, aligning with the study's dataset where a significant proportion of papers exhibit no AI-generated content. This structure accounts for cases where AI scores or plagiarism scores in research papers may be censored at zero, meaning that the underlying value could be negative but is not observed due to the nature of the scoring system.

Aside from Linear Tobit Model, the Quadratic Tobit Model, which extends the Linear Tobit Model by incorporating a squared term for time to capture potential nonlinear trends in AI score trajectories, and the Piecewise Tobit Model, which incorporates breakpoints in the time trend that allows for different growth rates across distinct periods, was also included in the study.

Lastly, the Tobit Model is implemented using the *AER* package in R for regression and *ggplot2* for visualization.

2.3.4 Latent Growth Mixture Model

Before exploring the Latent Growth Mixture Model (LGMM), it is important to first understand its foundational model, which is the Latent Growth Model (LGM). LGM is a statistical model composed of two latent components: the level (intercept) and the slope of growth, which consists of the mean value of the intercept and slope, and individual random variation of these two latent components [25].

However, LGM only captures AI score over time on a single growth model population, which is the reciprocal of Latent Growth Mixture Model (LGMM) as it allows two or more subpopulations. LGMM solves the homogeneity problem of LGM. Thus, it was adopted as it allows for identifying distinct unobserved subgroups, each following its own trajectory. This was essential given the diverse adoption rates of Generative AI across fields, as some disciplines may exhibit early, rapid growth while others show slower uptake.

LGMM supposes that the linear LGM for each subpopulation denoted by c follows the form:

$$y_{agg, w_{jt}}^c = \eta_{0j}^c + \eta_{1j}^c \cdot \lambda_t + \epsilon_{jt}^c \quad (2.12)$$

where $y_{agg, w_{jt}}$ is the vector of aggregated AI scores, ζ_{jt} is the random variables that configure each individual growth, η_j is the identified slope of latent components, ϵ_{jt} is

the measurement error, λ_t is the time score at time t ($t = 0, 1, 2, \dots, T - 1$), and c is the latent class.

However, the LGMM only outputs class growth models, thus the incorporation of conditions which are termed as time-varying covariate helps in uncovering underlying trends in discipline-level growth models. With the incorporation of Time-Varying Covariate (TVC), in this study, Plagiarism Score, the LGMM equation becomes:

$$y_{agg,wjt}^c = \eta_{0j}^c + \eta_{1j}^c \cdot \lambda_t + \beta_t^c \cdot x_{jt} + \varepsilon_{jt}^c \quad (2.13)$$

Moving forward, the researchers utilized two approaches in LGMM: the Quadratic LGMM and Piecewise LGMM, both with incorporation of Tobit Models for excess zero data [26].

Lastly, the LGMM is implemented in R using the *AER* package for Tobit regression, *ggplot2* for visualizing results, and *lcmm* package for LGMM.

3. Results and Discussion

This section provides an overview of Generative AI (GenAI) usage in research papers that was authored by undergraduate and graduate students at Bulacan State University (BuSU) from 2013 to 2023. It includes descriptive statistics, trend analysis, latent growth mixture model, and projections of AI Score evolution behavior beyond 2023.

3.1 Descriptive Analysis of GenAI Usage

This section focuses on the distribution of the number of research papers produced per year and per discipline, which highlights the trends in submission of research papers and GenAI usage over time. Additionally, the proportion of AI-flagged papers is also identified, which provides insights as to how the detection of AI-generated content has evolved.

Figure 1 shows the trend of research paper submissions over the years. The figure shows a steady increase from 2013 to 2019 and was followed by a sharp decline in 2020.

A trend of upward trajectory occurred from 2022 and 2023, indicating a partial recovery in research output post-pandemic.

Figure 2 shows the number of research papers per discipline, with Technology (T), Social Science (H), and Education (L) highlighting their dominance. The variation in research output across disciplines suggests differences in research activity levels.

Figure 1. Distribution of Research Papers Authored by Students per Year

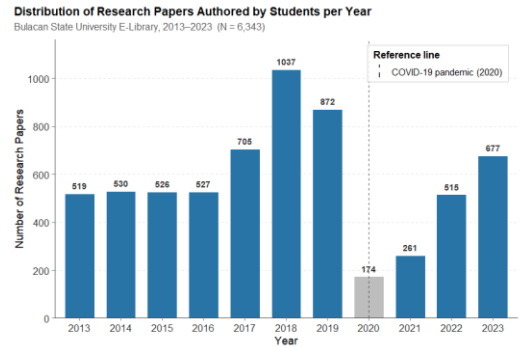


Figure 2. Distribution of Research Papers Authored by Students per Discipline

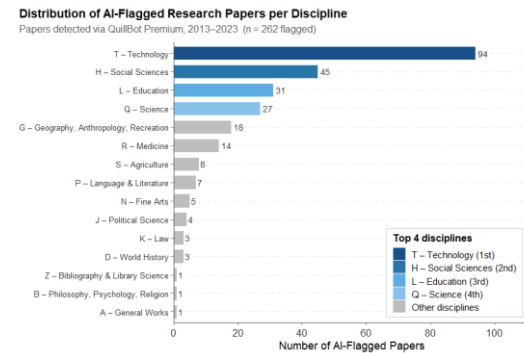


Figure 3 shows the upward trend in AI-flagged papers over time, having minimal detection of AI content before 2020 indicating limited usage, whereas the rapid increase in recent years suggests a growing usage of GenAI tools in research writing.

Figure 4 shows the concentration of AI-flagged papers per discipline, with Technology (T), Social Science (H), and Education (L) having the most numbers of papers flagged as AI-generated.

Figure 3. Distribution of AI-Flagged Papers per Year

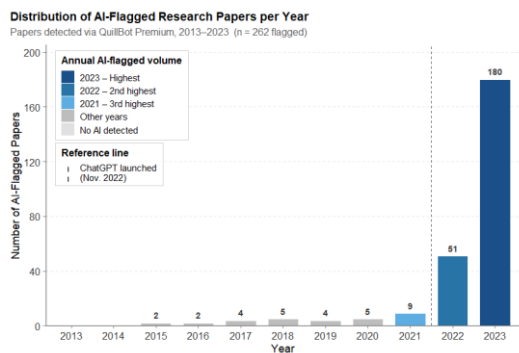
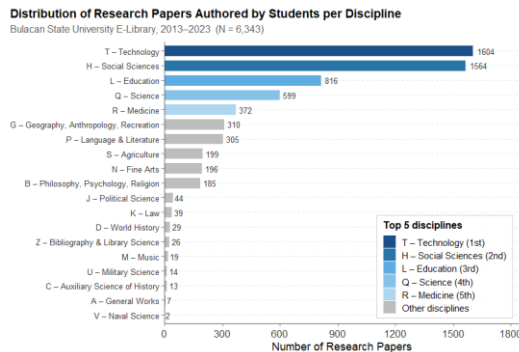


Figure 4. Distribution of AI-Flagged Papers per Discipline



3.2 Trend Analysis of GenAI Usage

This section presents the findings from the Zero-Inflated Beta (ZIB) Model and the Tobit Model, which were used to analyze the trends in GenAI usage over time. This serves as the starting point for examining how GenAI usage has evolved, considering different statistical approaches to account for variations in the dataset.

Table 1. Model Evaluation and Comparison of ZIB and Tobit Model

Model	AIC	BIC	R ²	RMSE
Individual-level Trend Analysis				
Quadratic Tobit*	1626.858	1653.878	0.3176	2.2105
Linear Tobit	1658.482	1678.748	0.3035	2.6289
Piecewise Tobit	1658.542	1685.563	0.3043	2.5422
ZIB	1876.697	1903.717	0.0901	0.4022

Discipline-level Trend Analysis				
Quadratic Tobit*	155.2694	171.9811	0.3515	0.5529
Tobit	156.6828	170.0521	0.3363	0.6532
Piecewise Tobit	158.5848	175.2964	0.3367	0.6673
ZIB	188.5403	205.252	0.3479	0.2414

Note: *Models marked with an asterisk (*) indicate the optimal model based on overall goodness-of-fit criteria.

Table 1 compares the performance of various models in analyzing GenAI usage trends at both individual and discipline levels using AIC, BIC, R², and RMSE. At the individual level, the Quadratic Tobit model shows the strongest performance, with the lowest AIC (1626.858), lowest BIC (1653.878), and highest McFadden’s Pseudo-R² (0.3176), indicating a good balance between model complexity and explanatory power. This R² falls within the moderate fit range (0.2–0.4), suggesting reasonable explanatory capability. At the discipline level, the Quadratic Tobit again performs best, with the lowest AIC (155.2694) and highest Pseudo-R² (0.3515), confirming its reliability in capturing trends across disciplines.

It is important to note that McFadden’s pseudo-R² is known to yield substantially lower values than the ordinary least squares coefficient of determination for equivalent levels of fit [29], and in censored regression models such as the Tobit, the partially unobserved nature of the dependent variable inherently constrains the proportion of variance that any model can account for in the observed data [23, 24]. Values in this range are therefore regarded as indicative of adequate fit in the censored regression literature [29], particularly when the model’s primary objective is characterizing longitudinal trend behavior rather than maximizing variance explanation.

Moreover, when comparing model performance, it is crucial to recognize that RMSE values between Tobit and ZIB models operate on fundamentally different scales

and thus cannot be directly compared. Tobit predicts on a latent scale (approx. -3 to 1), while ZIB uses a 0–1 observed AI score scale. Since RMSE is scale-dependent, comparisons are only meaningful within Tobit model variants.

Lastly, the Tobit model's RMSE (2.2105 individual, 0.5529 discipline) remains reasonable for long-term trend analysis despite being higher in absolute terms. Given these considerations, the Quadratic Tobit model is the best model for analyzing GenAI usage trends.

Table 2. Quadratic Tobit Model Estimates for AI Score

Predictor	Estimate	Std. Error	z-value	p-value
Individual-level				
(Intercept)	-2.4858	0.23615	-10.526	<0.001
Time	-0.1535	0.06340	-2.422	0.0154
Time ²	0.0347	0.00507	6.844	<0.001
Log(scale)	-0.1137	0.05343	-2.128	0.0333
Discipline-level				
(Intercept)	-0.7803	0.20645	-3.780	<0.001
Time	0.0043	0.06463	0.066	0.047
Time ²	0.0107	0.00531	2.015	0.044
Plagiarism Score	0.0312	0.03225	0.967	0.333
Log(scale)	-0.8940	0.11918	-7.502	<0.001

Table 2 presents the Quadratic Tobit Model estimates for AI Score at the individual-level, with its model expressed as $y_{it}^* = -2.4858 - 0.1535\text{Time} + 0.0347\text{Time}^2$, and at the discipline-level expressed as $y_{agg,wt}^* = -0.7803 + 0.0043\text{Time} + 0.0107\text{Time}^2 - 0.8940(\text{Plagiarism Score})$. At both levels, the model shows statistically significant upward trend at $p < 0.05$ for time coefficients. However, plagiarism scores demonstrated no significant influence at either level ($p = .333$). Lastly, the log scale estimates indicate moderate variability in adoption rates across papers and disciplines.

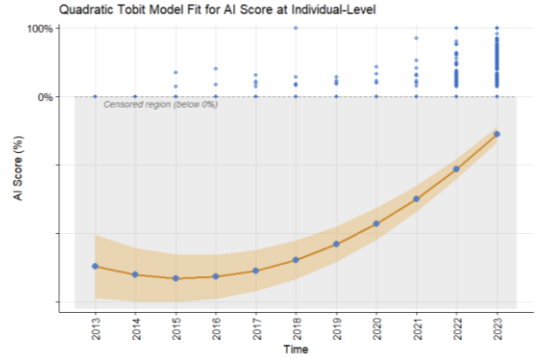


Figure 5. Quadratic Tobit Model Fit for AI Score at Individual-Level

Note: Blue dots represent observed AI scores, the orange curve represents the predicted AI trend, the shaded orange region represents the 95% confidence interval, and the shaded gray region indicates censored values treated as 0%.

Figure 5 presents the Quadratic Tobit regression fit for AI Score at the individual level from 2013 to 2023. The results indicate an accelerating trend in GenAI usage, but the persistence of low AI scores suggests that many research papers still do not incorporate AI-generated content. By 2023, the computed expected AI score is $y_{it}^* = -0.55$, which will be censored and reported as 0%, highlighting that a large portion of individual research papers still have no GenAI, despite an accelerating upward trend.

The accelerating yet still limited adoption of GenAI in research papers, with many scores censored at 0%, aligns with prior studies using AI detection tools. Akram [11] observed a gradual but uneven rise in AI-generated content in arXiv submissions after 2022, with computer science showing the highest adoption while other fields lagged. Similarly, Cheng et al. [10] found increased LLM usage in biomedical preprints following ChatGPT’s release, though findings still reflected largely human-like content. Kobak et al. [7] also noted that AI-generated text detection remains difficult due to hybrid human-AI writing patterns, which explain the persistence of low AI scores in many papers. Overall, these studies support the model’s finding of accelerating yet limited GenAI usage at the individual level.

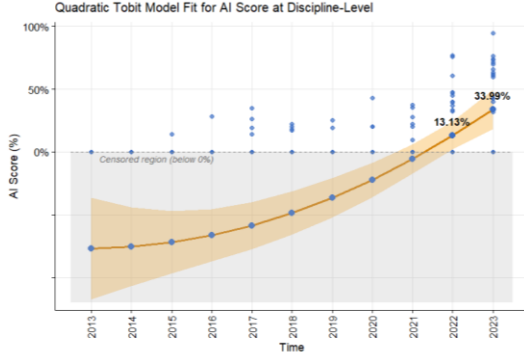


Figure 6. Quadratic Tobit Model Fit for AI Score at Discipline-Level

Note: Blue dots represent observed AI scores, the orange curve represents the predicted AI trend, the shaded orange region represents the 95% confidence interval, and the shaded gray region indicates censored values treated as 0%.

Figure 6 presents the Quadratic Tobit regression fit for AI Score at the discipline level from 2013 to 2023. The results indicate that GenAI usage is gradually accelerating across disciplines. By 2023, the expected AI score is $y_{agg,wt}^* = 0.3339$ or 33.99%, reflecting moderate but selective GenAI usage across disciplines.

The model’s projected discipline-level AI score of 33.99% aligns closely with recent empirical studies on GenAI usage while accounting for methodological differences across approaches. Liang et al. [9], using a mixture model, found that 17.5% of

Computer Science abstracts and 15.3% of introductions contained LLM-modified content by early 2024. Although the projected score is higher due to differences in methodology and broader disciplinary coverage, both studies highlight the accelerating trajectory of GenAI usage. This trend is further supported by another study [10], which analyzed preprint platforms and found rapid adoption in fields such as biomedicine, reaching up to 30%. Another study [7] also noted that AI detection accuracy varies across disciplines due to differences in writing style, being more standardized in technical fields but limited in notation-heavy fields such as mathematics. Collectively, these findings provide strong external validation for the model’s insight that GenAI usage is accelerating across disciplines, with clear field-specific variations in usage and detectability.

3.3 Latent Growth Mixture Model of GenAI Usage

Following the comprehensive discussion of trend analysis using Tobit models, this section presents the Latent Growth Mixture Model (LGMM) analysis, which explores the latent classes of GenAI usage across disciplines and their corresponding trajectories.

Table 3. Goodness-of-fit Statistics of GenAI Usage Latent Class for Quadratic and Piecewise

Class	Parameter	AIC	BIC	Entropy	Log-likelihood	p-value
Quadratic Latent Growth Mixture Model with Tobit						
1	8	1687.960	1695.516	1.0000	-835.9800	
2	13	1662.726	1675.004	0.9548	-818.3631	<.001
3*	18	1660.891	1673.891	0.9570	-815.4455	<.001
4	23	1675.472	1697.194	0.9132	-814.7358	.461
Piecewise Latent Growth Mixture Model with Tobit						
1	11	1742.169	1752.557	1.0000	-860.0843	
2	16	1742.118	1757.229	0.9254	-855.0588	.003
3	21	1749.454	1769.288	0.9057	-853.7272	.037
4	26	1758.052	1782.607	0.8694	-853.0260	.461

Note: The class marked with an asterisk (*) represents the optimal latent class solution and corresponding model structure based on overall goodness-of-fit indices, entropy, and model interpretability.

Table 3 compares the model fit statistics for the Quadratic and Piecewise LGMMs. Based on the goodness-of-fit indicators, the Quadratic LGMM consistently demonstrates lower AIC and BIC values, higher entropy, and better log-likelihood, indicating superior model fit over the Piecewise LGMM. Specifically, the three-class Quadratic LGMM presents the best overall fit, with an AIC of 1660.891, BIC of 1673.891, entropy of 0.9570, and log-likelihood of -815.4455, compared to the higher AIC (1749.454), BIC (1769.288), and lower entropy (0.9057) of the Piecewise LGMM.

These findings suggest that the quadratic trajectory best captures the latent structure of GenAI usage patterns and strongly indicate that the most appropriate latent structure consists of three distinct classes or trend trajectories, reflecting varying discipline-level patterns of GenAI adoption over time. With that, the succeeding tables further examine these latent classes through the fixed effects of the class-membership model and the longitudinal growth trajectories of each class over time.

Table 4. Parameter Estimates of the Quadratic LGMM Across Latent Classes

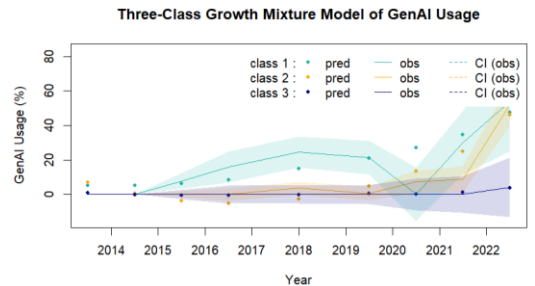
Predictor	Estimate	Std. Error	Wald	p-value
Intercept (Class 1)	5.1288	6.2988	0.814	<.001
Intercept (Class 2)	6.9328	2.5132	2.759	<.001
Intercept (Class 3)	0.9450	3.4687	0.272	<.001
Time (Class 1)	-0.5226	2.9679	-0.176	<.001
Time (Class 2)	-7.8621	1.2369	-6.356	<.001
Time (Class 3)	-1.1297	1.6732	-0.675	<.001
Time ² (Class 1)	0.5254	0.2878	1.826	<.001
Time ² (Class 2)	1.2557	0.1205	10.422	<.001
Time ² (Class 3)	0.1458	0.1658	0.879	<.001
Plagiarism Score	0.0713	0.0693	1.029	.3034

Table 4 presents the fixed effects for GenAI usage over time across three latent classes. Intercept values reflect initial usage: Class 2 starts highest ($\beta = 6.9328$, $p < .001$), followed by Class 1 ($\beta = 5.1288$, $p < .001$), while Class 3 starts lowest ($\beta = 0.9450$, $p < .001$), indicating varying baseline adoption levels across disciplines.

Linear time effects show negative trends in all classes: Class 2 has the steepest decline ($\beta = -7.8621$, $p < .001$), Class 3 a moderate decrease ($\beta = -1.1297$, $p < .001$), and Class 1 a slight drop ($\beta = -0.5226$, $p < .001$). Quadratic time effects reveal U-shaped trajectories in Classes 1 and 2, with Class 2 showing the strongest rebound ($\beta = 1.2557$, $p < .001$). Lastly, the plagiarism score ($\beta = 0.0713$, $p = .3034$) is not a significant predictor, which shows that the rise in GenAI usage is likely driven by other academic or technological factors, not plagiarism concerns.

Overall, these results highlight distinct GenAI usage patterns across the three latent classes. Class 2 exhibits an initial drop followed by a significant increase in GenAI usage, while Class 1 follows a similar but milder pattern. Class 3 shows a more stable trend with lower overall GenAI usage. These variations suggest that different disciplines have unique trajectories over time.

Figure 7. Three-Class Growth Mixture Model of GenAI Usage



Note: Based on the observed trend behaviors and the Innovation Diffusion Theory [27], Class 1 represents Innovators and Early Adopters, Class 2 represents the Early and Late Majority, and Class 3 represents Laggards. Blue dots represent observed AI scores, solid lines represent predicted trends, and shaded regions represent the corresponding confidence intervals.

Figure 7 represents the culmination of the Quadratic Latent Growth Mixture Model findings, illustrating the three latent (unobserved) classes of GenAI usage across disciplines. These classes align closely with the stages of adoption outlined in the Diffusion of Innovations [27], capturing distinct usage patterns shaped by historical trends, technological advancements, and the external influence of the COVID-19 pandemic. The researchers identify these three latent classes as Innovators and Early Adopters, Early and Late Majority, and Laggards, reflecting the typical diffusion trajectory of innovations within social systems.

To start with, Class 1 mirrors the Innovators and Early Adopters stages of IDT [27], comprising disciplines that integrated GenAI into their research and academic practices before the pandemic and the widespread rise of AI tools. These disciplines likely possessed stronger technological foundations or research alignment with AI applications, enabling early experimentation with tools such as paraphrasing software and grammar checkers before advanced AI systems became mainstream.

Next, Class 2 consists of disciplines that adopted GenAI after the pandemic, aligning with the rapid advancement and accessibility of AI-powered tools and large language models (LLMs) like ChatGPT. Corresponding to the Early Majority and Late Majority in IDT [27], these disciplines likely adopted GenAI after observing its success among earlier adopters, with the pandemic accelerating adoption by reducing complexity barriers and increasing the practicality of AI tools for research workflows.

Lastly, Class 3 represents disciplines that, despite advancements in AI and the transformative effects of the pandemic, have integrated GenAI minimally or not at all. Aligning with the Laggards stage of IDT

[27], these disciplines may reflect resistance to change, low awareness, structural constraints, or perceived incompatibility of GenAI with traditional methods.

By identifying these three distinct classes, the model provides a deeper understanding of how GenAI usage varies across disciplines, highlighting that while some fields were early users of GenAI, others adopted it as AI technologies became more mainstream, whereas some remain hesitant toward AI-driven research advancements.

Table 5. Class-Membership of Disciplines based on GenAI Usage

Class of GenAI Usage	Discipline	Proportion
1 – Innovators and Early Adopters	L – Education	31.58%
	T – Technology	
	B – Philosophy, Psychology, Religion	
	D – World History	
	G – Geography, Anthropology, Recreation	
2 – Early and Late Majority	H – Social Sciences	57.89%
	J – Political Sciences	
	K – Law	
	N – Fine Arts	
	P – Language and Literature	
	Q – Science	
	R – Medicine	
	S – Agriculture	
	A – General Works	
3 – Laggards	C – Auxiliary Science in History	10.53%
	M – Music	
	U – Military Science	
	V – Naval Science	
	Z – Bibliography, Library Science, Information Resources	

Table 5 provides the membership of disciplines in the three latent classes. Notably, Discipline L (Education) and T (Technology) are identified as innovators and early adopters, demonstrating an early

and steady usage of GenAI before the advancements of Large Language Models (LLMs) such as ChatGPT. In addition, the majority of disciplines in Class 2 show an early and late acceleration of GenAI usage pattern, while some disciplines are identified as laggards whose GenAI usage remains minimal.

Furthermore, Table 6 reveals the distribution of disciplines across the three latent classes: Innovators and Early Adopters (31.58%), Early and Late Majority (57.89%), and Laggards (10.53%). While Rogers' Innovation Diffusion Theory [27] showed a near-symmetrical distribution of adopters, this study's front-loaded curve, marked by a higher proportion of Innovators/Early Adopters (31.58%) and a smaller Laggard group (10.53%), suggests an accelerated diffusion of GenAI in academia. This deviation may reflect the unique influence of the pandemic and rapid advancements in GenAI, which compressed adoption timelines for many disciplines (e.g., Education [L] and Technology [T]). The dominance of the Early/Late Majority (57.89%) aligns with Rogers' expectation that most adopters belong to the middle stages, though their faster-than-typical uptake highlights the critical role of observability such as visible success in peer disciplines, and reduced complexity (e.g., user-friendly AI tools) in driving adoption [27].

It is important to note that these classifications are based on research disciplines as categorized by the Library of Congress classification system, which organizes fields of study by subject matter. This classification does not directly correspond to specific colleges or academic programs. For example, the discipline of Education does not strictly refer to the College of Education but rather to research conducted in the field of education across multiple academic units. Lastly, each discipline is associated with multiple

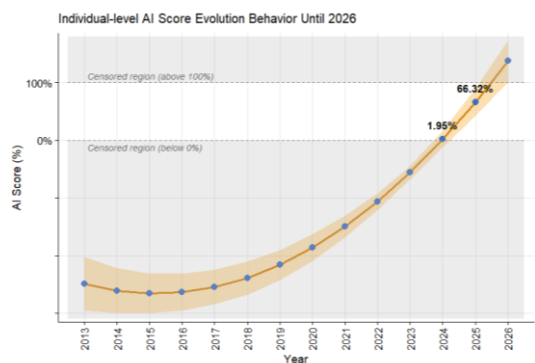
colleges and programs. This ensures that the analysis remains focused on research fields rather than individual colleges, avoiding any unintended harm for specific colleges and programs.

3.4 AI Score Evolution Behavior Beyond 2023

This section presents the behavior of GenAI usage trend beyond 2023 at both the individual and discipline levels, based on the Quadratic Tobit model and Quadratic LGMM. While the Quadratic Tobit model shows moderate fit (McFadden's pseudo- $R^2 \approx 0.32\text{--}0.35$), and reasonable RMSE and MAE given the censored nature of the data, these projections are not intended for precise prediction but to illustrate macro-level growth behavior of trajectories.

This modeling approach is aligned with previous study who applied a Tobit model to short time series data with 10 quarters to understand trend behavior and projection beyond that 10-quarter period [28]. Similarly, the aim here is to capture the shape of GenAI usage trends until reaching their censoring limit. With that, the figures and tables below show how GenAI usage trends behave until reaching saturation.

Figure 8. Individual-level AI Score Evolution Behavior Until 2026



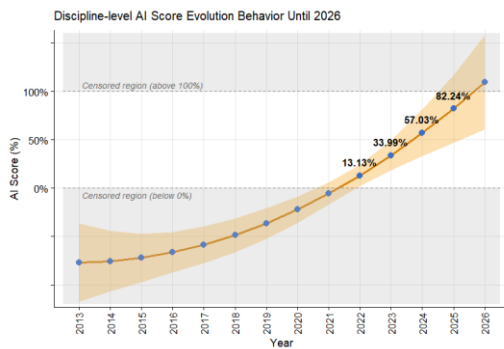
Note: The orange curve represents the predicted AI score with 95% CI. Blue dots are observed individual AI scores. The shaded gray region indicates censored values, where projected values below 0% are reported as 0% and values above 100% are reported as 100%.

Figure 8 shows the trend behavior of individual-level GenAI usage from 2013 to

2026, with a 95% confidence interval. The curve follows a quadratic trajectory, with the latent AI score remaining entirely below 0% throughout 2013–2023, meaning all individual-level scores during this period are left-censored and recorded as 0% [23, 24]. The model only crosses into the observable range ($y^* > 0\%$) beginning in 2024, yielding a projected AI score of 1.95%. This is followed by a dramatic surge to 66.32% in 2025, before reaching the right-censoring limit of 100% by 2026.

The prolonged censoring at 0% through the observed period reflects that GenAI usage at the individual paper level remained latent rather than directly measurable, while the steep post-2023 acceleration highlights the transformative impact of widely accessible large language models [5, 1]. This trajectory captures the full momentum of adoption as it rapidly approaches saturation within just three years of becoming observable [27].

Figure 9. Discipline-level AI Score Evolution Behavior Until 2026



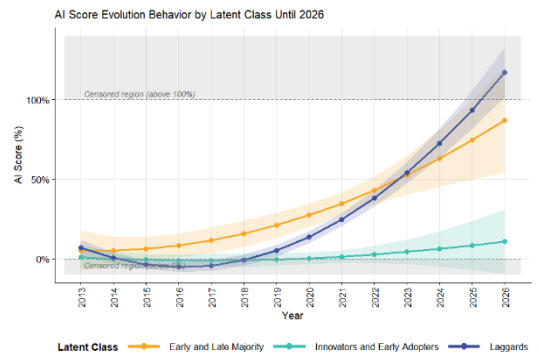
Note: The orange curve represents the predicted AI score with 95% CI. Blue dots are observed individual AI scores. The shaded gray region indicates censored values, where projected values below 0% are reported as 0% and values above 100% are reported as 100%.

Figure 9 presents the projected GenAI usage at the discipline level from 2013 to 2026, also following a quadratic trajectory. At the discipline level, the aggregated AI score first enters the observable range ($y^* > 0\%$) in 2022, registering 13.13%, a threshold that coincides with the widespread post-launch adoption of ChatGPT [1, 4]. By 2023,

the score rises meaningfully to 33.99%, confirming that discipline-level GenAI usage had already reached a moderate and detectable level within the study period [11, 9]. The trajectory then accelerates sharply beyond 2023, reaching 57.03% in 2024 and 82.24% in 2025, before being right-censored at 100% by 2026.

Compared to the individual level, discipline-level AI usage becomes observable two years earlier (2022 vs. 2024), reflecting the aggregation effect: even disciplines with sparse individual AI-flagged papers contribute a detectable pooled signal [26]. The steepness of the post-2022 curve mirrors the inflection observed at the individual level and collectively highlights a macro-level shift toward widespread GenAI usage across all academic fields within a compressed timeframe [3, 1].

Figure 10. AI Score Evolution Behavior by Latent Class Until 2026



Note: Solid lines are predicted class trajectories with 95% CI bands. The shaded gray region indicates censored values, where projected values below 0% are reported as 0% and values above 100% are reported as 100%. See Appendix B for complete projected values per class.

Figure 10 shows the behavior of GenAI usage trends across the three latent classes from the Quadratic LGMM, emphasizing their distinct trajectories. The complete year-by-year projected AI scores for each class are presented in Appendix B.

Class 1 (Innovators and Early Adopters) shows consistent and uninterrupted observable AI usage from as early as 2013, beginning at 5.25% and rising steadily through 11.65% in 2017, 21.11% in

2019, and 34.77% in 2021, reaching 52.65% by the end of the observed period in 2023. Projections indicate continued but moderating growth, reaching 63.15% in 2024, 74.72% in 2025, and 87.33% by 2026, reflecting a gradual approach to saturation [27].

Class 2 (Early and Late Majority) exhibits a more volatile trajectory: it shows initial usage in 2013 (7.13%) and 2014 (5.27%), before dropping to 0% from 2015 through 2018, years in which GenAI tools had not yet become mainstream [2, 5]. It resurfaces in 2019 at 5.16%, then accelerates rapidly following the rise of LLMs [5], reaching 24.60% in 2021, 38.08% in 2022, and notably overtaking Class 1 in 2023 with 54.08%. This surge continues post-2023, with projections of 72.59% in 2024 and 93.61% in 2025, before reaching the right-censoring limit of 100% by 2026.

Class 3 (Laggards) remains almost entirely censored at 0% through 2019, with only marginal observable usage emerging from 2020 (0.38%), gradually increasing to 1.43% in 2021, 2.78% in 2022, and 4.42% in 2023. Projections suggest continued but slow growth, reaching 11.09% by 2026, far below the other two classes.

These trajectories reveal a key divergence: Class 2, despite its slower and interrupted start, ultimately surpasses Class 1 by 2023 and achieves full adoption earlier, driven by the transformative impact of ChatGPT and similar LLMs that lowered barriers to GenAI use across mainstream disciplines [1, 4]. The reversal of rank between Class 1 and Class 2 suggests that early adoption does not guarantee continued leadership when a disruptive technology undergoes a step-change in accessibility [27]. Meanwhile, the persistently low and slowly growing usage in Class 3 points to structural and disciplinary constraints, whether methodological incompatibility, limited awareness, or institutional resistance [27], that continue to impede meaningful GenAI

integration even as surrounding fields rapidly converge toward full adoption [7].

4. Conclusion and Recommendations

This section shows the summary of key findings, conclusions drawn from the analysis, and practical recommendations for addressing GenAI usage in academic research. The insights offer guidance for institutions to develop effective policies while upholding research integrity.

4.1 Conclusion

The findings of this study reveal significant and accelerating trends in GenAI usage in academic research from 2013 to 2023. At the individual level, all AI scores remained latent and censored at 0% throughout the observed period, with the model projecting that observable usage first emerges in 2024 at 1.95%, then surges to 66.32% in 2025 before reaching full adoption by 2026. At the discipline level, GenAI usage became detectable as early as 2022 at 13.13%, rising to 33.99% by 2023 and projected to reach 100% by 2026, reflecting that aggregated discipline-level signals capture adoption earlier than individual-level scores.

Moreover, across three distinct latent trajectories identified by the Quadratic LGMM, Class 1 (Innovators and Early Adopters) showed consistent usage from 2013 (5.25%) with a projected ceiling of 87.33% by 2026, while Class 2 (Early and Late Majority), despite dropping to 0% from 2015 to 2018, which overtook Class 1 by 2023 (54.08% vs. 52.65%) and is projected to reach 100% by 2026. Class 3 (Laggards) remains substantially below both, reaching only 4.42% in 2023 and a projected 11.09% by 2026.

The findings show no significant correlation between plagiarism levels and GenAI usage, suggesting that the rise in AI adoption is driven by broader academic and technological factors rather than by plagiarism-related behaviors. In addition,

comparisons of models confirm that the Quadratic Tobit approach best captures both individual and discipline-level trends, and the Quadratic LGMM provides deeper insight into heterogeneous adoption patterns across disciplines.

Notably, it is important to interpret these projections with appropriate caution. The Quadratic Tobit model and Quadratic LGMM were trained on data spanning 2013 to 2023, a period in which meaningful non-zero AI scores only began to emerge consistently from 2020 onward. This means the post-2023 projections, particularly the rapid approach to censoring limits at 100%, are extrapolated from effectively three to four years of observable signal. While the quadratic trajectory is statistically well-supported within the observed period, the steep acceleration beyond 2023 should be understood as an illustration of macro-level trend behavior under the model's current trajectory rather than a precise empirical forecast. External factors such as institutional policy changes, evolving detection tool sensitivity, shifts in research norms, or the emergence of new AI technologies could meaningfully alter the actual trajectory. Replication of this study with updated data from 2024 onward is therefore strongly encouraged to validate or recalibrate these projections.

Overall, the evidence points to the near-complete integration of GenAI in academic research within just a few years, highlighting the urgent need for evidence-based and discipline-sensitive institutional policies and responses.

4.2 Recommendations

To promote responsible AI usage while safeguarding academic integrity, the following recommendations are proposed:

1. Given the sharp and accelerating rise in GenAI usage evidenced by the projections, academic institutions are urged to establish or strengthen clear

institutional policies governing appropriate GenAI use in research and academic writing. These policies should be proactively communicated and consistently enforced across all disciplines and academic levels, with particular attention to fields showing the steepest adoption trajectories.

2. Institutions are encouraged to adopt standardized AI detection tools, such as Turnitin, Copyleaks, or other validated platforms, and make them broadly accessible to faculty and students for academic use. Standardized detection supports transparent and consistent assessment of AI involvement in submitted work, and helps build a shared understanding of what constitutes appropriate GenAI assistance across the institution.
3. Rather than applying a uniform AI score threshold across all research fields, institutions should consider implementing discipline-specific thresholds informed by latent class analyses such as those presented here. Fields in the Early and Late Majority class that are trending toward full adoption may warrant different standards than Laggard disciplines with persistently low usage. Enforcing a discipline-sensitive thresholds improve fairness, reduce misclassification of legitimate AI-assisted work, and allow policies to evolve in step with actual adoption patterns.
4. Future studies are encouraged to incorporate additional factors such as technological influences, demographic characteristics, and researchers' perceptions of GenAI. Replication of this study in other institutions is encouraged to develop context-specific AI integration guidelines.

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Appendix

Appendix A. Scope and Delimitation

This study examines Generative AI (GenAI) usage in research papers stored in the Bulacan State University E-Library, aiming to analyze the trends and growth trajectories of GenAI usage from 2013 to 2023. The analysis includes theses, capstone projects, and dissertations authored by graduate and undergraduate students. A full population approach was employed with certain exclusions, specifically papers without abstracts, abstracts containing fewer than 80 words, and papers written in Filipino, as AI detection tools cannot consistently process such papers accurately. The study was conducted during the 2024 to 2025 academic year, covering data retrieval, processing, and analysis.

Furthermore, the projections presented for years beyond 2023 are derived from a model trained on a period during which non-zero AI scores were only consistently observable from approximately 2020, limiting the available signal for post-observation extrapolation to roughly three

data points per level. These projections are therefore intended to characterize trend direction and saturation behavior rather than serve as definitive predictive estimates, and should be revisited as new data becomes available.

Moreover, the study solely focuses on empirical trend analysis, excluding external factors such as technological advancements, academic policy changes, and shifts in educational practices that may influence GenAI usage. Due to its accessibility and capability to process large volumes of text, QuillBot Premium was utilized as a third-party AI and plagiarism detection tool. A full population approach with exclusions was applied by considering only papers that met the study criteria. However, because of financial and practical constraints, plagiarism scores were analyzed only at the discipline level as a time-varying covariate. The study also acknowledges the limitations of detection tools, including false positives and false negatives, which were addressed using Zero-Inflated Beta (ZIB) and Tobit models to account for excess zero values and

misclassification errors. Lastly, the study has several delimitations to provide privacy and uphold ethical standards. In order to ensure confidentiality, no personal information about the research authors, advisers, or panelists will be disclosed or revealed. The researchers emphasize that this study does not intend to make false accusations or harm any institution or individual but aims to provide a neutral, data-driven analysis of GenAI usage trends in academic papers. To prevent any unintended harm among individuals or colleges within Bulacan State University, the data was aggregated into disciplines based on the Library of Congress Classification System rather than individual courses or colleges to ensure anonymity. To further ensure data privacy and responsible research practices in strict adherence to Republic Act No. 10173, or the Data Privacy Act of 2012, all data collected will be permanently deleted five years after the study's completion. The results are intended to inform both academia and stakeholders regarding the evolving role of GenAI technologies in academic research.

Appendix B. AI Score Evolution Behavior by Latent Class Until 2026 Based on the Quadratic LGMM

Year	Class 1 Projected AI Score	Class 1 AI Score (%)	Class 2 Projected AI Score	Class 2 AI Score (%)	Class 3 Projected AI Score	Class 3 AI Score (%)
2013	0.0525	5.25%	0.0713	7.13%	0.0114	1.14%
2014	0.0529	5.29%	0.0527	5.27%	0.0016	0.16%
2015	0.0638	6.38%	-0.0035	0.00%*	-0.0053	0.00%*
2016	0.0849	8.49%	-0.0051	0.00%*	-0.0093	0.00%*
2017	0.1165	11.65%	-0.0042	0.00%*	-0.0104	0.00%*
2018	0.1585	15.85%	-0.0078	0.00%*	-0.0085	0.00%*
2019	0.2111	21.11%	0.0516	5.16%	-0.0384	0.00%*
2020	0.2742	27.42%	0.1363	13.63%	0.0038	0.38%
2021	0.3477	34.77%	0.2460	24.60%	0.0143	1.43%
2022	0.4318	43.18%	0.3808	38.08%	0.0278	2.78%
2023	0.5265	52.65%	0.5408	54.08%	0.0442	4.42%
2024	0.6315	63.15%	0.7259	72.59%	0.0635	6.35%
2025	0.7472	74.72%	0.9361	93.61%	0.0857	8.57%
2026	0.8733	87.33%	1.1713	100.00%**	0.1109	11.09%

Note: Values marked with * are left-censored at 0, indicating that negative projections are treated as 0% AI usage. Values marked with ** are right-censored at 100%, meaning AI usage is capped at 100%.

Appendix C. Algorithm of Classic Tobit Regression

- 1: Initialize parameters x_{it}, β with starting values
 - 2: For each dataset (individual and aggregated):
 - 3: If individual-level:
 - 4: Set $y = y_{it}$ and $X = x_{it}$
 - 5: If aggregated-level:
 - 6: Set $y = y_{agg, wjt}$ and $X = \beta * X_{agg, wjt}$
 - 7: Handle censoring:
 - 8: If $y \leq \tau$, mark as censored
 - 9: If $y > \tau$, mark as uncensored
 - 10: Estimate likelihood function:
 - 11: Uncensored: $y = X\beta + \varepsilon$
 - 12: Censored: Integrate over truncated normal distribution
 - 14: Optimize parameters β using MLE
 - 15: Compute fitted values $\hat{y} = X\beta$
 - 16: Return estimated parameters and fitted values
-

Appendix D. Algorithm of Quadratic Tobit Regression

- 1: Initialize parameters x_{it}, β with starting values
 - 2: For each dataset (individual and aggregated):
 - 3: If individual-level:
 - 4: Set $y = y_{it}$ and $X = x_{it}$
 - 5: If aggregated-level:
 - 6: Set $y = y_{agg, wjt}$ and $X = \beta * X_{agg, wjt}$
 - 7: Transform predictors by adding quadratic terms:
 - 8: $X^* = [X, X^2]$
 - 9: Handle censoring:
 - 10: If $y \leq \tau$, mark as censored
 - 11: If $y > \tau$, mark as uncensored
 - 12: Estimate likelihood function:
 - 13: If uncensored: $y = X^*\beta + \varepsilon$
 - 14: If censored: Integrate over truncated normal distribution
 - 15: Optimize parameters β using MLE
 - 16: Compute fitted values $\hat{y} = X^*\beta$
 - 17: Return estimated parameters and fitted values
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Appendix E. Algorithm of Piecewise Tobit Regression

- 1: Initialize parameters $x_{it}, \beta_1, \beta_2, \beta_3$ with starting values
 - 2: For each dataset (individual and aggregated):
 - 3: If individual-level:
 - 4: Set $y = y_{it}$ and $X = x_{it}$
 - 5: If aggregated-level:
 - 6: Set $y = y_{agg, wjt}$ and $X = \beta_3 * X_{agg, wjt}$
 - 7: Segment data based on thresholds:
 - 8: If $y \leq \tau_1$: Assign to Segment 1
 - 9: If $\tau_1 < y \leq \tau_2$: Assign to Segment 2
 - 10: If $y > \tau_2$: Assign to Segment 3
 - 11: Estimate separate Tobit models for each segment
 - 12: Fit individual regression models for each segment
 - 15: Handle censoring appropriately in each segment
 - 15: Optimize parameters $\beta_1, \beta_2, \beta_3$ using MLE
 - 16: Compute fitted values $\hat{y}$ for each segment
 - 17: Return estimated parameters and fitted values
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Appendix F. Algorithm of Quadratic Tobit Latent Growth Mixture Model (LGMM)

- 1: Initialize the following for each class c :
 - 2: Class membership probabilities π_c
 - 3: Growth parameters $\beta_{0c}, \beta_{1c}, \beta_{2c}$
 - 4: For each class c in C :
 - 5: Define the Tobit model with quadratic growth:
 - 6: $y_{agg, wjt} = \beta_{0c} + \beta_{1c}t + \beta_{2c}t^2 + \varepsilon$
 - 7: Handle censoring at threshold τ :
 - 8: If $y_{agg, wjt} \leq \tau$, treat as censored
 - 8: If $y_{agg, wjt} > \tau$, treat as uncensored
 - 10: Estimate class-specific Tobit models using Maximum Likelihood Estimation (MLE)
 - 11: Compute posterior probabilities:
 - 12: $P(c|y_{agg, wjt})$ using the Expectation-Maximization (EM) algorithm
 - 13: Update class membership probabilities π_c
 - 14: Assign each observation to the most probable class.
 - 15: Repeat until convergence criteria are met.
 - 16: Return estimated parameters and class assignments.
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Appendix G. Algorithm of Piecewise Tobit Latent Growth Mixture Model (LGMM)

- 1: Initialize the following for each class c :
 - 2: Class membership probabilities π_c
 - 3: Growth parameters $\beta_{01c}, \beta_{11c}, \beta_{02c}, \beta_{12c}$
 - 4: For each class c in C :
 - 5: Define the Piecewise Tobit model:
 - 6: If $t \leq t^*$: $y_{agg, wjt} = \beta_{01c} + \beta_{11c}t + \varepsilon$
 - 7: If $t > t^*$: $y_{agg, wjt} = \beta_{02c} + \beta_{12c}t + \varepsilon$
 - 8: Handle censoring at threshold τ :
 - 9: If $y_{agg, wjt} \leq \tau$, treat as censored
 - 10: If $y_{agg, wjt} > \tau$, treat as uncensored
 - 11: Estimate class-specific Tobit models using Maximum Likelihood Estimation (MLE)
 - 12: Compute posterior probabilities:
 - 13: $P(c|y_{agg, wjt})$ using the Expectation-Maximization (EM) algorithm
 - 14: Update class membership probabilities π_c
 - 15: Assign each observation to the most probable class.
 - 16: Repeat until convergence criteria are met.
 - 17: Return estimated parameters and class assignments.
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